Linear Relationships Test

Section A: Identify Linear Relationships

Look at the table below. Is the relationship linear? Explain your reasoning.
x: 1, 2, 3, 4

y: 3, 5, 7, 9

The table shows values of x and y. Determine whether the pattern is linear.
x: 0, 2, 4, 6

y: 10, 8, 6, 4

A relationship is shown in the table. Decide if it is linear.
x: 1, 3, 5, 7

y: 4, 9, 14, 19

Section B: Describe the Rule

Use the table to find the rule connecting x and y.
x: 2, 4, 6, 8

y: 7, 11, 15, 19

Write a rule for the relationship shown.
x: 0, 1, 2, 3

y: 5, 8, 11, 14

Determine the rule for this linear pattern.
x: 3, 5, 7, 9

y: 12, 16, 20, 24

Section C: Use the Rule to Find Missing Values

A linear rule is $y = 4x + 1$. Find y when $x = 6$.

A relationship follows the rule $y = 2x - 3$. Complete the table.
x: 1, 4, 7, 10

7. y: __, __, __, __
8. The rule is $y = 5x$. What is the value of y when $x = 9$?

Section D: Create Your Own Table or Graph

10. Create a table of values for the rule $y = 3x + 2$ using at least four x-values.
11. Draw a simple graph that represents the rule $y = x + 4$.
12. Make a table and sketch the graph for the rule $y = 2x - 1$.

Section E: Worded Linear Relationship Problems

13. A taxi charges a $6 starting fee plus $2 per kilometre. Write a rule for the total cost C based on kilometres k.
14. A video game store rents games for $4 per day. Write a rule for the total cost T for d days.
15. A plant grows 3 cm every week. If its starting height is 10 cm, write a rule for its height H after w weeks.

End Test